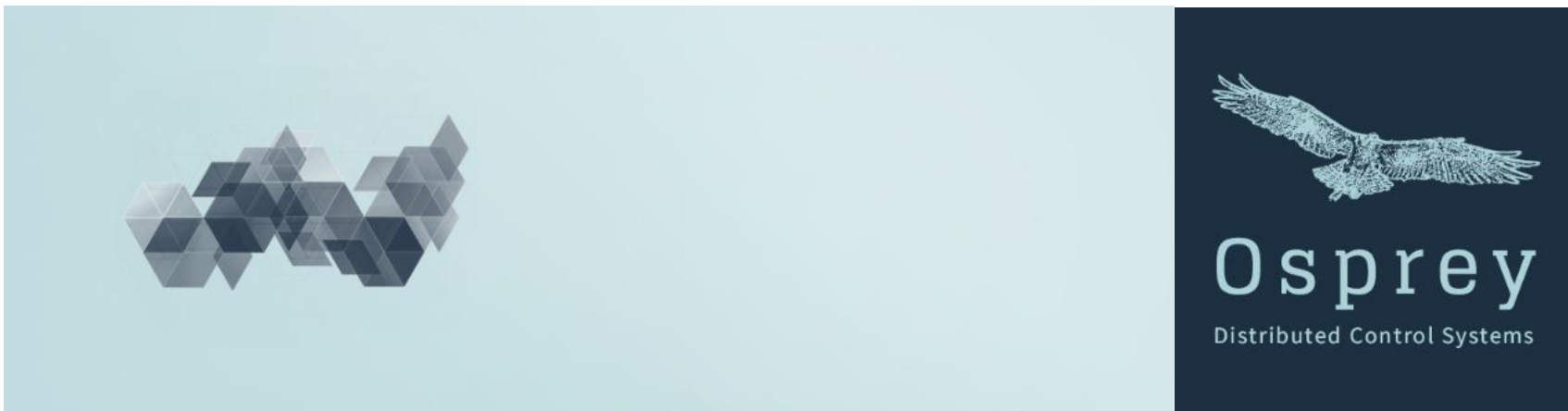




# Machine Learning Data Platform

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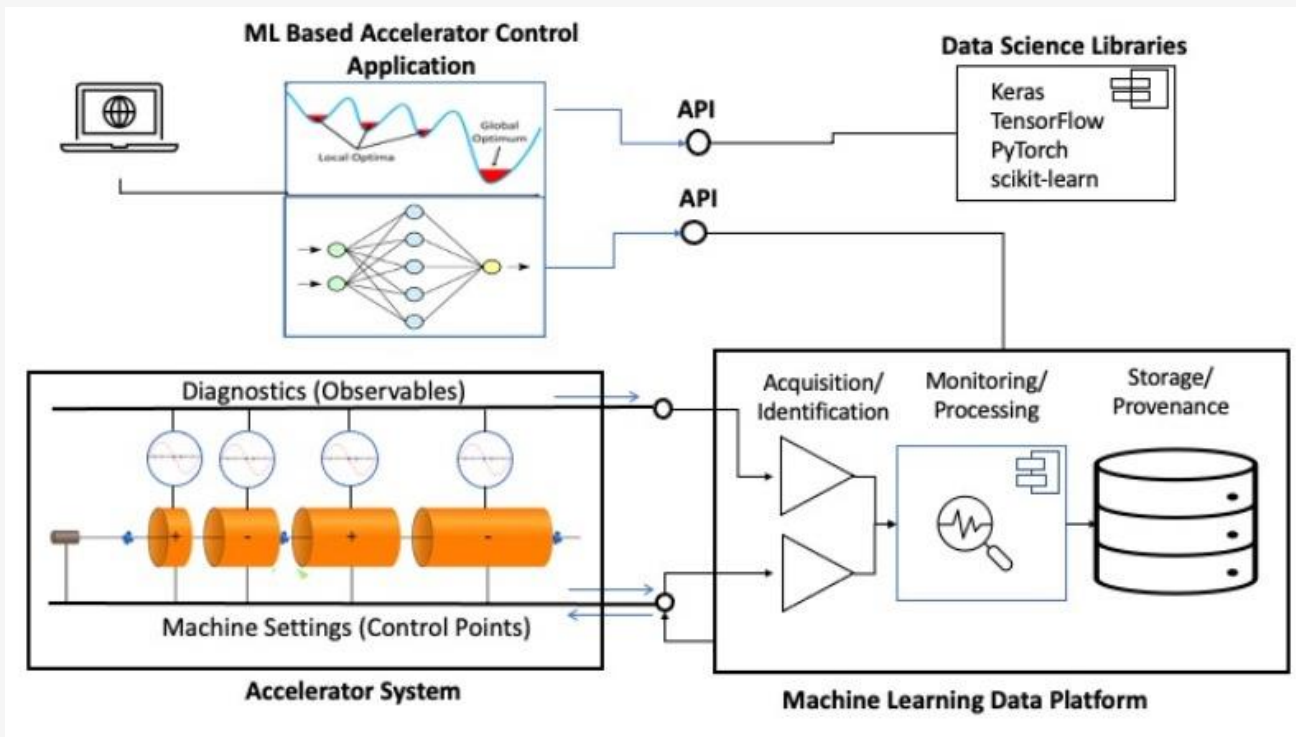
# Topics

Machine Learning Data Platform  
Overview

APIs: Ingestion, Metadata / Annotation,  
Query, and Subscription

Development Plans

Learn More



The MLDP provides tools for 1) building an annotated archive of PV time-series data and 2) using the archive to build and operate data science applications.

- Provides gRPC APIs for
  - PV time-series data ingestion
  - metadata and annotation
  - data retrieval
  - data / event subscription
- APIs are handled by scalable Java services using MongoDB to manage the archive
- The ecosystem includes
  - Python and Java API clients for building ML applications
  - desktop and web GUIs
  - administrative tools and configuration examples

# MongoDB Archive – Particle Accelerator Data

Process Variables (Columns)

Timestamps (Rows)

Time Samples

|                     | PV1               | PV2              | PV3                    | PV4               |
|---------------------|-------------------|------------------|------------------------|-------------------|
| Timestamp           | Beam Energy (GEV) | Magnet Field (T) | RF Cavity Voltage (MV) | Beam Current (ma) |
|                     | PV1               | PV2              | PV3                    | PV4               |
| 2023-07-15 12:00:00 | 4.20              | 1.50             | 12.5                   | 250.4             |
| 2023-07-15 12:01:00 | 4.18              | 1.55             | 12.7                   | 248.9             |
| 2023-07-15 12:01:00 | 4.22              | 1.52             | 12.5                   | 251.1             |
| 2023-07-15 12:03:00 | 4.19              | 1.54             | 12.6                   | 249.7             |
| 2023-07-15 12:04:00 | 4.21              | 1.56             | 12.8                   | 252.0             |

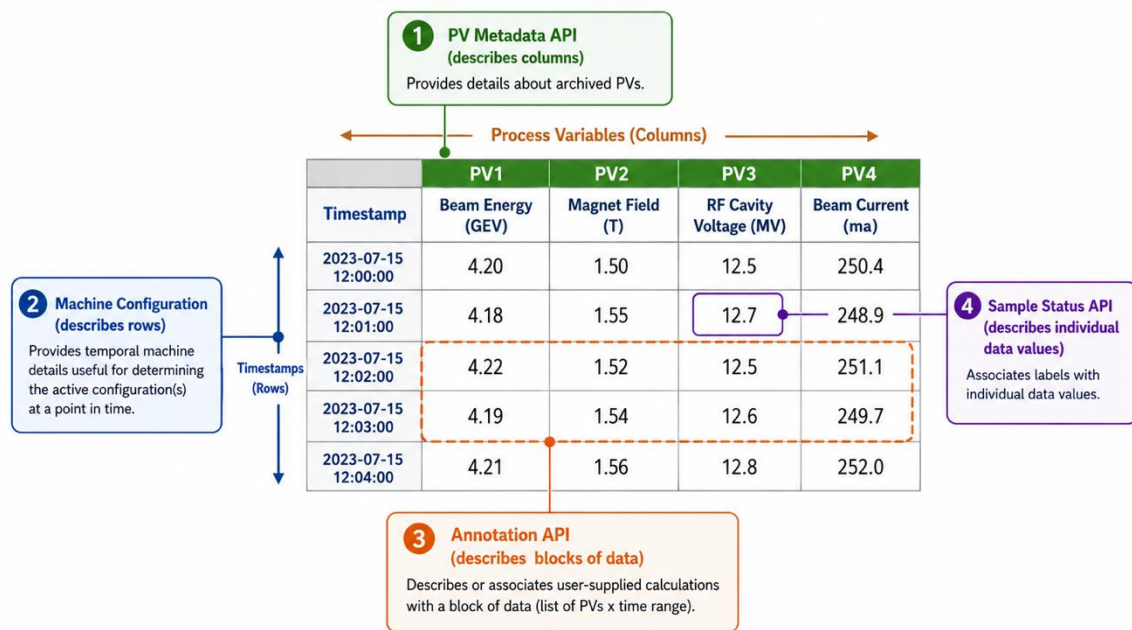
PV Data Values

## PV Time-Series Data Ingestion API

- Provides optimized structures for ingesting a variety of data types including scalars, multi-dimensional arrays, structures, images, and serialized data
- Utilizes streaming API methods and supports static and dynamic load balancer configurations for maximum performance
- Includes mechanisms for specifying data provenance and ingestion metadata
- Supports both continuous and batch ingestion

## MLDP Archive and Metadata APIs

The MLDP Archive can be thought of as a giant spreadsheet of accelerator data.  
The 4 MLDP metadata APIs describe the columns, rows, blocks of data, and individual data values in that spreadsheet.



Metadata and annotation APIs cover 4 different dimensions of archive

### PV Metadata API

Answers “what are the properties for PV name BPMS:GUNB:314:X”?

```
DEVICE=BPMS:GUNB:314
ELEMENT=BPM1B      TYPE=MONI
Z= -9.555017        S= 0.489650
AREA=GUNB
BEAMPATHS={SC_HXR, SC_DIAG0, SC_HXR}
```

### Temporal Machine Configuration API

Answers “what was the machine configuration at 2-Feb-2026 18:04:01”?

```
PATH=CU_HXR      E=14.6 (GeV);
RATE=10000 (Hz)  MODE=09;
DEST=CXI          EXP=CXI_3443
```

### Annotation / Calculations API

Describes a block of data and allows linking user-supplied calculations to archive data.

### Sample Status API

Tags individual data values, e.g.,

```
Time=141412342134.13412342 secs.nanos
PV=BPMS:GUNB:314:X
Value=2.4
Tag=SUSPECT
Domain=RMS fit
```

# Data Retrieval and Export API

- Provides query interface that leverages archive metadata for filtering queries by time, PV metadata, temporal machine configuration, and sample status
  - SQL analogy

```
SELECT <PV time-series data>
FROM <the MLDP archive>
WHERE
    <Time Range Criteria> AND
    <PV Selection Criteria> AND
    <Machine Configuration Criteria> AND
    <Sample Status Criteria>
```
  - e.g., "Get me all *\*good\** BPM data from area L3 between time1 and time2 when we were running beam and the experiment was cxi-Q4r4234."
- Incorporates paging, streaming, and multiple query result formats to support a wide range of use cases
- Includes utilities for exporting data to common file formats
- Python and Java clients provide high level tools for building ML applications using the query API

# Data / Event Subscription API

- Enables clients to subscribe to live PV data from the ingestion stream
- Provides a mechanism for receiving notification of live data events from the ingestion stream with a configurable window of data around the trigger time
- We are investigating approaches for running user code as a plugin, and mechanisms for publishing data using frameworks like Kafka

# MLDP Development Plans

- Currently supporting deployment at SLAC
- Development priorities for upcoming year
  - ingestion API and database optimizations for handling shared timestamps
  - configurable strategies for improved distribution of data in Mongo shards
  - age-based archival of data to external storage while maintaining Mongo indexes
  - enhance query API to support query by PV data value
  - Python client library enhancements for building ML applications
  - improvements to data subscription frameworks
  - support for large atomic data values that exceed the 16MB Mongo object limit
  - tools for ingesting data from EPICS environment
  - improved metrics capture and observability
- On hold / looking for funding
  - JWT authentication / role-based authorization / LDAP integration
  - data ownership and sharing mechanisms
  - enhancements to desktop and web GUI applications
  - Java client library catch up to v2 API enhancements

# For Additional Information

The primary GitHub repo for the Machine Learning Data Platform project is <https://github.com/osprey-dcs/data-platform> . It includes project overview, status, road map, quick start guide, installation instructions, and links to secondary repos and additional project documents.

Osprey project team

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